

FIFA WORLD CUP EVOLUTION

Evaluating Tournament Success from 1994 to 2026

An end-to-end SQL and Power BI analytics project examining tournament growth, sporting performance, stadium utilisation, financial performance and global audience reach.

THE QUESTION THAT STARTED IN A SITTING ROOM

This project did not begin as a classroom exercise or a pre-written case study. It began as a **sitting-room argument while the 2026 FIFA World Cup was being played**. People were calling the tournament poor, disappointing or over-commercialised, while the headline numbers appeared to show record scale, packed stadiums and enormous global attention. Rather than continue arguing from impressions, the debate became a data problem: **what does the evidence actually say?**

Central analytical question

How has the modern FIFA World Cup evolved since USA 1994, and was the 2026 tournament genuinely successful for both FIFA and football fans?

PostgreSQL • SQL • Data Cleaning • Data Validation • Power BI • DAX • Data Modelling •
Data Visualisation

1. Executive Summary

The project compares nine modern FIFA World Cups from 1994 through 2026. The period was narrowed from the original 1930-2026 concept because modern editions provide stronger and more comparable attendance, commercial and audience evidence, while USA 1994 provides a natural North American benchmark for 2026.

The workflow was built from the ground up. A relational PostgreSQL database was assembled from multiple official and historical sources; tournament, match, venue, attendance, sporting, financial and audience data were standardised and linked; SQL was used to audit the data and create analysis-ready views; and the validated model was then taken into Power BI.

Headline 2026 results

Metric	2026 result	Interpretation
Teams	48	+50.0% vs 2022
Matches	104	+62.5% vs 2022
Total attendance	~7.0M	+100.1% vs 2022
Average attendance	~65K	+23.1% vs 2022
Capacity utilisation	99.74%	+3.2% vs 2022
Goals	308	2.96 goals per match
World Cup cost	\$3.76bn	2026 budget/provisional basis
Revenue / budget	\$8.91bn	2026 budget/provisional basis
Projected surplus	\$5.16bn	Revenue/budget less cost
Audience / engagement	5.20bn	Reported engagement measure
Digital / video views	20.00bn	Reported digital/video measure
Social impressions	30.00bn	Reported social measure

Executive verdict: the measurable evidence does not support describing 2026 as a poor World Cup overall. It was an exceptional success in scale, stadium demand, commercial potential and digital reach, while sporting output remained strong. Fan affordability, travel and visa concerns remain important limitations because they are not captured as consistently as the core performance metrics.

2. Project Scope and Analytical Framework

The formal scope defined the project as a comparative analysis of commercial growth, sporting performance, stadium attendance, global reach and fan experience. It recognised the central tension that a World Cup can be commercially successful for FIFA without necessarily being equally successful for supporters.

Why 1994-2026?

- USA 1994 provides a natural North American benchmark for the 2026 three-country edition.
- The period captures expansion from 24 teams and 52 matches to 48 teams and 104 matches.
- It includes single-country, dual-country and three-country hosting models.
- Commercial, ticketing and audience information is more available in the modern era.
- The narrower period reduces reliance on estimates and improves comparability.

Tournament editions in scope

Year	Host(s)	Analytical relevance
1994	United States	North American benchmark
1998	France	Expansion to 32 teams
2002	Korea Republic / Japan	First dual-hosted Asian World Cup
2006	Germany	Modern stadium/fan benchmark
2010	South Africa	First World Cup in Africa
2014	Brazil	Football-market comparison
2018	Russia	Modern broadcast/commercial benchmark
2022	Qatar	Compact recent benchmark
2026	Canada / Mexico / United States	First 48-team, 104-match, three-country edition

3. Building the Relational Database

The analysis was not based on one clean CSV. Information came from FIFA annual and financial reports, audience reports, official stadium-capacity releases, match and attendance records and 2026 tournament updates. These sources had different grains and definitions, so the first major task was to build a consistent relational structure.

Table / layer	Purpose
tournaments	One record per World Cup edition and the central tournament key.
matches	Match-level records linked to tournament and venue.
venues	Stadium identity and venue descriptors.
tournament_venues	Tournament-era venue capacity, avoiding the assumption that capacity is timeless.
match_attendance	Attendance and available-capacity denominator by match.
sporting layer	Goals, draws, winning margins and discipline.
financial performance	Revenue categories, cost and actual/budget/provisional status.
audience performance	Audience, TV, digital/video and social measures with measurement labels.
SQL views	Analysis-ready layers for Power BI and consistent business logic.

The design separated entities that change at different grains. Stadium capacity was especially important: the tournament_venues layer allowed the model to store the capacity relevant to a specific World Cup rather than assigning one permanent capacity to a stadium.

4. Data Cleaning, Standardisation and Validation

Cleaning was treated as analytical work, not cosmetic preparation. The greatest risk was mixing numbers that looked similar but represented different concepts, reporting bases or measurement definitions.

Key work performed

- **Keys and relationships:** tournament, match and venue identifiers were standardised so joins relied on keys rather than text names.
- **Historical capacity correction:** an earlier 1994-2022 capacity mapping was replaced after it produced incorrect denominators; corrected tournament-era capacities were propagated into match_attendance.
- **Coverage validation:** SQL counted matches with and without capacity by tournament.
- **Occupancy audit:** matches with attendance above listed capacity were surfaced for review instead of silently deleted.
- **Missing data:** unavailable historical metrics remained NULL/blank rather than being filled with unsupported estimates.
- **Financial status:** actual, budget and provisional figures were kept distinguishable; 2026 was not treated as final audited World Cup actuals.
- **Audience definitions:** final audience, reach, video views, impressions and engagement were kept separate.
- **Comparability:** digital visuals show only years where a defensible metric exists rather than inventing a complete historical series.

The capacity-denominator correction was one of the most important quality-control steps. After updating tournament-era capacities, the project reran coverage and utilisation audits so an incorrect denominator could not contaminate the capacity-utilisation KPI.

5. SQL Analysis and Business Logic

PostgreSQL served as both the storage layer and the first analytical layer. JOINS connected tournaments, matches, venues and attendance; GROUP BY and aggregations created tournament-level measures; and validation queries tested calculated outputs before they were visualised.

Representative capacity-utilisation formula

$$\text{SUM(attendance)} * 100.0 / \text{SUM(available_capacity)}$$

SQL work included JOINS, SUM/COUNT/AVG aggregations, GROUP BY year, ROUND, arithmetic calculations, coverage checks, exception audits, UPDATE statements for corrected source mappings, and analysis-ready views including the financial-performance layer.

6. Power BI Model and DAX

Tournament year became the main filtering dimension in Power BI. DAX measures used SELECTEDVALUE to return the selected edition and comparison measures identified the latest earlier tournament with a valid value.

Important DAX behaviour

- The first valid observation displays **Baseline** instead of a false percentage change.
- Previous-edition comparisons search for a nonblank prior value, which is essential when a metric is not available every four years.
- Cost increases are formatted differently from positive increases in revenue, surplus, audience or reach.
- Display measures bridge historical actuals with 2026 budget/provisional fields while preserving the reporting caveat.
- Digital and social measures return BLANK for years without evidence.

7. Final Power BI Report

The finished report contains four interconnected pages with a tournament-year slicer, a custom left navigation bar, trophy branding, consistent green-and-gold styling and active-page highlighting.

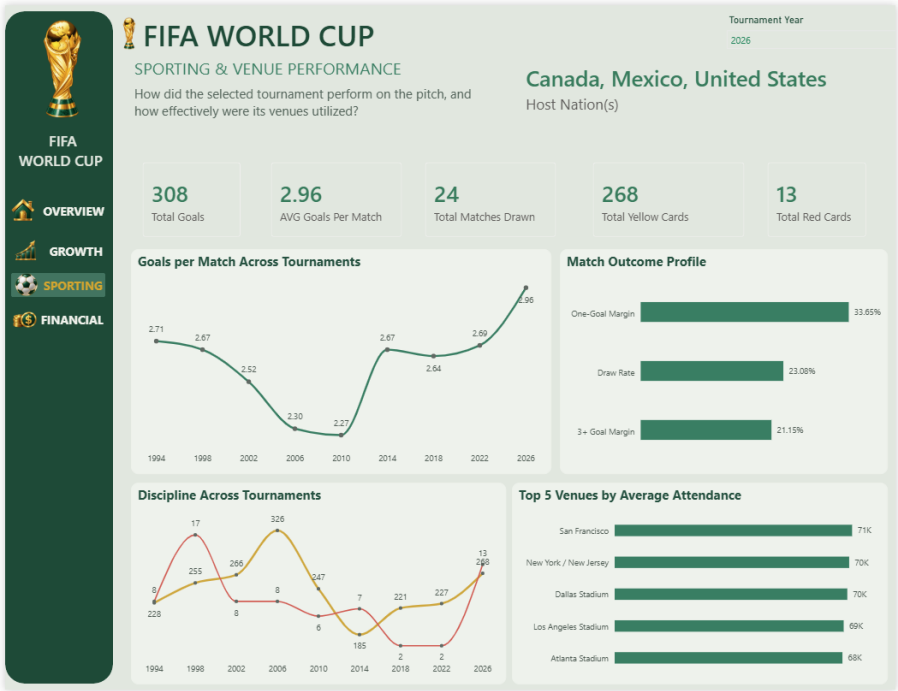
Dashboard 1 - Growth & Global Impact



Dashboard 2 - Growth & Comparisons



Dashboard 3 - Sporting & Venue Performance



Dashboard 4 - Financial & Global Reach



8. Key Findings - Growth and Global Impact

The first page establishes the structural story. The World Cup expanded from 24 teams and 52 matches in 1994 to 48 teams and 104 matches in 2026.

Indicator	Historical context	2026	Finding
Teams	24 in 1994; 32 from 1998-2022	48	Largest field in tournament history
Matches	52 in 1994; 64 from 1998-2022	104	62.5% above 2022
Total attendance	Earlier editions mostly ~2.7M-3.6M	~7.0M	Approximately double 2022
Average attendance	Historical benchmark ~52K	~65K	26.98% above benchmark
Capacity utilisation	Historical benchmark ~94.5%	99.74%	Near-full despite expansion

The attendance record cannot be dismissed as a simple consequence of more matches. Total attendance benefits from 104 fixtures, but average attendance also rose materially and stadium utilisation approached 100%. Scale and genuine demand increased together.

9. Key Findings - Historical Comparison

2026 KPI	Result	Interpretation
Teams growth vs previous	+50.00%	32 to 48 teams
Matches growth vs previous	+62.50%	64 to 104 matches
Attendance growth vs previous	+100.09%	Total attendance approximately doubled
Average attendance vs historical	+26.98%	Growth was not only more fixtures
Capacity utilisation vs historical	+5.27%	2026 outperformed the historical benchmark

This comparison is central to the project. Record totals alone can mislead when tournament size changes. By pairing totals with per-match and utilisation measures, the dashboard shows that 2026's growth was supported by strong demand, not merely a larger schedule.

10. Key Findings - Sporting Performance

Measure	2026 result	Interpretation
Goals	308	High total output across 104 matches
Goals per match	2.96	Highest displayed value in the 1994-2026 trend
Drawn matches	24	Competitive outcomes remained common
One-goal margin	33.65%	Many matches were decided narrowly
Draw rate	23.08%	Meaningful competitive balance
3+ goal margin	21.15%	Large wins existed but did not dominate
Yellow / red cards	268 / 13	Discipline remained measurable historically

11. Key Findings - Financial Performance and Global Reach

The financial page combines commercial performance with global audience and digital engagement. Because financial reporting bases and digital metrics are not perfectly comparable across eras, the dashboard labels 2026 as budget/provisional and restricts digital comparisons to years with available data.

2026 metric	Dashboard value	Status / interpretation
World Cup cost	\$3.76bn	2026 cost/investment basis
Revenue / budget	\$8.91bn	Budget/provisional basis used in model
Surplus / projected surplus	\$5.16bn	Calculated revenue/budget less cost
Audience reach / engagement	5.20bn	Reported engagement measure
Digital reach / video views	20.00bn	Reported digital/video measure
Social-media impressions	30.00bn	Reported social-impression measure

The revenue-mix visual shows broadcasting as a major commercial pillar, alongside marketing, hospitality/ticketing and licensing. The 2026 mix must be read with the reporting-basis caveat: the values are not all final audited World Cup-specific actuals.

The audience chart intentionally leaves older digital years blank. A blank means unavailable; zero would incorrectly imply measured absence.

12. Limitations and Responsible Interpretation

The project became stronger when it stopped forcing every originally proposed fan-experience metric into the final dashboard. Several series - historical resale prices, ticket-to-income ratios, visa-refusal effects and comparable supporter-satisfaction surveys - were not sufficiently complete for a defensible 1994-2026 comparison.

- **2026 financials:** the \$5.16bn figure is a projected analytical surplus, not certified accounting profit.
- **Digital metrics:** video views, impressions and engagement are not unique-person audience counts and should not be summed.
- **Historical audience:** measurement definitions changed across eras; final audience and cumulative reach are different constructs.
- **Capacity:** announced attendance and official seating capacity may use different definitions, so isolated values slightly above 100% require review.
- **Fan experience:** attendance and engagement demonstrate demand and reach but cannot prove affordability, easy travel, convenient visa access or universal satisfaction.
- **Causality:** the report identifies comparative patterns; it does not claim that expansion alone caused revenue, attendance or engagement growth.

Core methodological rule: missing data remains missing. Unavailable historical digital metrics were not turned into zeroes, and budget figures were not silently converted into actuals.

13. Answering the Sitting-Room Argument

Was the 2026 FIFA World Cup genuinely a poor tournament?

No - not on the measurable performance evidence covered by this project. The data supports classifying 2026 as a highly successful World Cup in structural scale, stadium demand, sporting output, commercial potential and global/digital reach.

The case is strongest in the stadium data. The competition expanded to 48 teams and 104 matches, yet average attendance rose to roughly 65,000 and capacity utilisation reached 99.74%. The record total attendance therefore was not produced by extra fixtures alone.

The sporting evidence also conflicts with a simple 'bigger but worse' narrative. The completed dataset records 308 goals and 2.96 goals per match, the highest value shown in the modern goals-per-match trend, while one-goal margins and draws remained substantial parts of the outcome profile.

Commercially, the available 2026 budget/provisional model points to a very large event: \$8.91bn in the revenue/budget measure against \$3.76bn in World Cup cost, producing a \$5.16bn projected analytical surplus. Digital engagement operated at a different scale from earlier eras, with 20bn video views and 30bn social impressions represented in the report.

The conclusion is not that every criticism was false. The final dashboard cannot conclusively settle affordability, visas, travel burden or subjective supporter experience because comparable historical evidence is incomplete. The defensible conclusion is therefore: **the available performance data strongly opposes the claim that 2026 was a poor tournament overall, while some fan-experience criticisms remain unresolved by these metrics.**

Final classification

Strong overall tournament success, with exceptional stadium, scale, commercial and engagement performance; fan-accessibility questions remain a qualified limitation rather than evidence of overall tournament failure.

14. Recommendations and Future Analytical Work

This is not a conventional business case requiring recommendations to FIFA management. Its main purpose is an evidence-based verdict. Still, the work suggests good analytical practice for future tournament evaluation.

Recommendation	Reason
Keep totals and per-match metrics together	Prevents expansion from being mistaken for stronger demand.
Publish comparable fan-accessibility indicators	Would balance commercial/stadium success with affordability, travel, visas and sentiment.
Separate actual, budget and provisional finance	Prevents projected success from being presented as audited realised performance.
Standardise audience definitions	Keeps unique reach, cumulative reach, views and impressions conceptually separate.
Preserve tournament-era venue capacity	Avoids incorrect utilisation denominators.
Refresh after final 2026 reporting	Replace provisional fields with audited values and rerun comparisons.

The most useful next analytical step is to refresh the PostgreSQL database when final audited 2026 financial and audience reports are available and then extend the fan-accessibility layer only where the evidence can be verified.

15. Selected Source Base

Source	Role
FIFA World Cup Evolution - Project Scope & Analytical Framework	Defines scope, question, hypotheses and verification rules.
FIFA Annual / Financial Reports	Revenue, cost, ticketing/hospitality and tournament-cycle context.
FIFA television / audience reports	Historical reach and audience methodology.
FIFA 2026 stadium-capacity releases	Venue capacity and occupancy validation.
FIFA 2026 digital/audience milestone releases	Attendance, occupancy and digital milestones.
PostgreSQL database and correction scripts	Relational integration, cleaning and validation.
Power BI model and DAX	KPIs, previous-edition baselines and dashboards.

End of report